Sadra Gholami

Data Science / Machine Learning

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PROFILE

Industrial Engineering undergraduate focused on data science, machine learning, operations research, and optimization. Interested in applying analytical and programming skills to practical problems, developing data-driven solutions, and contributing to collaborative technical teams.

EDUCATION

B.Sc. in Industrial Engineering | Shahid Bahonar University of Kerman | 2022 – 2026 | GPA: 16.26/20

TECHNICAL SKILLS

Advanced: Python, Operations Research, Exploratory Data Analysis, Object-Oriented Programming, Git/GitHub
Intermediate: Machine Learning, Data Structures & Algorithms, SQL, Feature Selection, Data Cleaning & Preprocessing, Data Visualization, Model Evaluation & Validation, Scientific Research
Beginner: AI Infrastructure

EXPERIENCE

Teaching Assistant | Shahid Bahonar University of Kerman | Feb 2024 – Jul 2026

- Linear Algebra (4 semesters)
- Machine Learning (2 semesters)
- Decision Analysis (1 semester)

Member | Industrial Engineering and Systems Scientific Association, Shahid Bahonar University of Kerman | Jun 2025 – Sep 2026

- Association Secretary
- Managing Editor, Sana Student Scientific Journal
- Secretary-General, Faculty of Engineering Associations

SELECTED PROJECTS

Maintenance Analytics for Hezaar Aluminum Industries

- Performed statistical and predictive analysis of electrical and mechanical maintenance data for the A6 machine; developed performance indicators.
- Tools: Python, Pandas, NumPy, Scikit-learn, Jupyter, Matplotlib, Seaborn | GitHub: github.com/gsadra/Maintenance-Analytics-for-Hezaar-Aluminum-Industries

Multidimensional Fitness Function-Based Heuristic for the Set Covering Problem

- Implemented an object-oriented Python heuristic algorithm for the Set Covering Problem, with configurable variables and parameters.
- Tools: Python, Dataclasses, Pytest | PyPI: pypi.org/project/mdfh-scp/

Cancer Diagnosis Prediction and Classification

- Analyzed clinical features and implemented KNN, SVM, and Decision Tree models; achieved an F1-score of 0.96.
- Tools: Pandas, Scikit-learn, Jupyter, Matplotlib, Seaborn

Customer Segmentation

- Analyzed retail customer data and applied K-Means and DBSCAN; examined cluster cores and the Elbow method.
- Tools: Pandas, Scikit-learn, Matplotlib, Seaborn

CERTIFICATIONS

- **CS50x: Introduction to Computer Science** — Harvard University
- **CS50P: Introduction to Programming with Python** — Harvard University
- **Machine Learning with Python** — Maktabkhooneh
- **Future Managers Professional Skills Bootcamp** — Iran Elites Network

LANGUAGES

Persian: Native | English: Advanced | Dutch: Beginner